

2020 PRIMES Expository Paper

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1 Introduction

This paper will be concerned with the Riemann zeta function and the derivation of the prime number theorem. Historically, Euclid first famously proved in 300 BC that there are infinitely many primes, but for centuries, it was not known how prime numbers were distributed. Legendre in 1808 calculates by hand the number of primes at most x , denoted $\pi(x)$, and the density of primes, $\frac{\pi(x)}{x}$, for x up to 1,000,000. He concluded numerically that $\pi(x) \simeq \frac{x}{\log x}$ as $x \rightarrow \infty$ which is the statement of the prime number theorem. Euler then worked with the function $\sum_{n=0}^{\infty} n^{-s}$ and showed that

$$\sum_{n=0}^{\infty} n^{-s} = \prod_p \frac{1}{1 - p^{-s}}.$$

Substituting $s = 1$ gives the harmonic series on the left which grows as $\log n$ and Euler heuristically wrote $\prod_p \frac{1}{1 - p^{-s}} = \log \infty$. Chebyshev showed in two papers in 1848 and 1850 that if $\frac{\pi(x)}{x/\log x}$ exists, it must be 1.

Later, Riemann wrote his famous thesis "On the Number of Primes Less Than a Given Magnitude" [8] where he extended $\zeta(s) = \sum_{n=0}^{\infty} n^{-s}$ analytically to the entire complex plane and use the logarithmic derivative

$$(\log(\zeta))' = \frac{\zeta'}{\zeta}(s) = - \sum_p \frac{p^{-s} \log p}{1 - p^{-s}}$$

and an integral formula that is later known as Perron's formula to derive a so-called explicit formula for primes, thus introducing complex analysis into number theory.

Finally, the details in Riemann's paper were made rigorous using Hadamard's factorization theorem [3][4], and Hadarmard [4] and De La Vallée Poussin [9][10] giving independent proofs in 1896 that $\zeta(s)$ had no zeros of the form $1 + it$, thus proving the prime number theorem. Our paper mainly discusses Riemann's paper but uses the Chebyshev functions instead of $\pi(x)$ directly for cleaner arguments.

Finally, we conclude with a discussion of p -adic numbers, Adèles, and Idèles which can be used to generalize analogues of the prime number theorem to algebraic number fields. The idea comes from the letter from Iwasawa to Dieudonné [5], but is restricted to the field of rational numbers only.

2 The Maximum Principle, by Ishita Goluguri

The content of this section mostly comes from [2].

Definition 1 (Holomorphic). *A complex valued function f defined on some region on the complex plane is holomorphic if it is complex differentiable in a neighborhood of every point in its domain.*

Theorem 2.1. *Every zero of a non-constant holomorphic function is isolated.*

Theorem 2.2. *Every holomorphic function is complex analytic, i.e., for any point of its domain, there is a neighbourhood in which the function can be represented a convergent power series.*

Theorem 2.3 (Cauchy Integral Formula). *Suppose f is holomorphic on some region that contains the disk $B(a, r)$. There holds the Cauchy Integral formula*

$$f(a) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z-a} dz,$$

where γ is any circle inside the disk $B(a, r)$, surrounding a and oriented counterclockwise.

Theorem 2.4 (Mean Value Theorem). *Let f be a holomorphic function on $B(a, R)$. For any $r < R$, $f(a) = \frac{1}{2\pi} \int_0^{2\pi} f(a + re^{i\theta}) d\theta$.*

Proof. This proof follows from the Cauchy Integral formula. The path γ can be expressed as $a + re^{i\theta}$ for $0 \leq \theta \leq 2\pi$. Thus,

$$\begin{aligned} f(a) &= \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z-a} dz \\ &= \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(a + re^{i\theta})}{(a + re^{i\theta}) - a} \frac{d}{d\theta}(a + re^{i\theta}) d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(a + re^{i\theta}) d\theta. \end{aligned}$$

□

Theorem 2.5 (Maximum Principle). *Let Ω be a domain, and let f be holomorphic on Ω .*

(A) *$|f(z)|$ cannot attain its maximum inside Ω unless f is constant.*

(B) *The real part of f cannot attain its maximum inside Ω unless f is a constant.*

Part A. Let $|f|$ attain its maximum on some $a \in \Omega$. Let $\overline{B}(a, r) \subset \Omega$ be a neighborhood of a . By assumption, $\max_{\theta} |f(a + re^{i\theta})| \leq |f(a)|$. By the Mean Value Theorem,

$$|f(a)| \leq \frac{1}{2\pi} \int_0^{2\pi} |f(a + re^{i\theta})| d\theta,$$

and again, by the assumption that f reaches its maximum over a ,

$$\frac{1}{2\pi} \int_0^{2\pi} |f(a + re^{i\theta})| d\theta \leq \max_{\theta} |f(a + re^{i\theta})|.$$

Thus,

$$\max_{\theta} |f(a + re^{i\theta})| \leq |f(a)| \leq \frac{1}{2\pi} \int_0^{2\pi} |f(a + re^{i\theta})| d\theta \leq \max_{\theta} |f(a + re^{i\theta})|$$

We can conclude that $\max_{\theta} |f(a + re^{i\theta})| \cong |f(a)|$ and $f(z) = f(a)$ in every neighborhood of a , so f is a constant function. $\square$

This principle is used again in the proof of the Borel Caratheodory Lemma, which is important in the proof of the Weierstrass Factorization Theorem.

Corollary 2.5.1. *From the Maximum Principle, we can conclude that*

$$\sup_{z \in \Omega} |f(z)| \leq \sup_{z \in \partial\Omega} |f(z)|$$

and

$$\sup_{z \in \Omega} \operatorname{Re}(f(z)) \leq \sup_{z \in \partial\Omega} \operatorname{Re}(f(z)).$$

Theorem 2.6 (Jensen's Inequality). *Suppose f is holomorphic on the whole complex plane and $f(0) = 1$. Let $M_f(R) = \max_{|z|=R} |f(z)|$. Let $N_f(t)$ be the number of zeros of f with norm $\leq t$ where a zero of multiplicity n is counted n times. Then*

$$\int_0^R \frac{N_f(t)}{t} dt \leq \log M_f(R).$$

Proof. To make this proof a little simpler, we can assume that R is not the norm of any of the zeroes. If it is, the theorem will follow anyway by continuity. Consider the disk $B(0, R)$ and let the zeroes of f inside this disk be $a_1, a_2, \dots, a_N$. We want to remove these zeroes but still keep the norm of $f(z)$ unchanged on this disk, so we define the function h where

$$h(z) = f(z) \left(\prod_{k=1}^N \frac{R(z - a_k)}{R^2 - \bar{a}_k z} \right)^{-1}$$

We know that $|f(z)| = |h(z)|$ for $|z| = R$. h is a holomorphic function on $B(0, R)$ and has no zeroes on the disk $B(0, R)$ because all of the zeroes of $f(z)$ have been canceled, so $\log(h)$ is a holomorphic function on the disk. Using the mean value theorem on the function $\log(h)$ on this disk and setting $a = 0$,

$$\log |h(0)| = \frac{1}{2\pi} \int_0^{2\pi} \log |h(Re^{i\theta})| d\theta.$$

Using the definition of $h(z)$, we see that

$$\log |h(z)| = \log |f(z)| - \sum_{k=1}^N \log \left| \frac{R(z - a_k)}{R^2 - \bar{a}_k z} \right|,$$

and

$$\log |h(0)| = \log |f(0)| - \sum_{k=1}^N \log \frac{|a_k|}{R},$$

and by definition of $h(z)$,

$$\log |h(0)| = - \sum_{k=1}^N \log \frac{|a_k|}{R} = \sum_{k=1}^N \log \frac{R}{|a_k|}.$$

Substituting,

$$\sum_{k=1}^N \log \frac{R}{|a_k|} = \frac{1}{2\pi} \int_0^{2\pi} \log |h(Re^{i\theta})| d\theta.$$

Using the fact that $|f(z)| = |h(z)|$ for $|z| = R$, we find

$$\sum_{k=1}^N \log \frac{R}{|a_k|} = \frac{1}{2\pi} \int_0^{2\pi} \log |h(Re^{i\theta})| d\theta.$$

The right-hand-side of the equation is bounded above by $M_f(R)$, while the left side is calculated as

$$\begin{aligned} \sum_{|a_k| < R} \log \frac{R}{|a_k|} &= \log \left(\frac{R^N}{|a_1||a_2|\dots|a_N|} \right) \\ &= \log \frac{|a_2|}{|a_1|} + 2 \log \frac{|a_3|}{|a_2|} + 3 \log \frac{|a_4|}{|a_3|} + \dots + N \log \frac{R}{|a_N|} \\ &= \int_{|a_1|}^{|a_2|} \frac{dt}{t} + \int_{|a_2|}^{|a_3|} \frac{2dt}{t} + \int_{|a_3|}^{|a_4|} \frac{3dt}{t} + \dots \\ &= \int_0^R \frac{N_f(t)}{t} dt. \end{aligned}$$

This concludes the proof. □

Theorem 2.7 (Borel-Carathéodory Lemma). *Suppose $f = u + iv$ is holomorphic on the whole complex plane. Suppose $u \leq A$ on $\partial B(0, R)$. Then*

$$|f^{(n)}(0)| \leq \frac{2n!}{R^n} (A - u(0))$$

Proof. Because $f(z)$ is an analytic function, we can write

$$f(z) = \sum_{n=1}^{\infty} c_n z^n, \quad c_n = \frac{f^{(n)}(0)}{n!} = |c_n| e^{i\theta_n}$$

using Taylor series expansion. As seen in the givens, $u(z) = \operatorname{Re}(f(z))$. Thus,

$$u(Re^{i\theta}) = \operatorname{Re}(f(Re^{i\theta})) = \operatorname{Re} \sum_{n=1}^{\infty} |c_n| e^{i\theta_n} R^n e^{in\theta} = \sum_{n=1}^{\infty} |c_n| R^n \cos(n\theta + \theta_n).$$

So

$$\int_0^{2\pi} [u(Re^{i\theta}) - u(0)] d\theta = \sum_{n=1}^{\infty} |c_n| R^n \int_0^{2\pi} \cos(n\theta + \theta_n) d\theta = 0,$$

$$\int_0^{2\pi} [u(Re^{i\theta}) - u(0)] \cos(n\theta + \theta_n) d\theta = \pi |c_n| R^n.$$

Adding these two equalities, we get

$$\int_0^{2\pi} [u(Re^{i\theta}) - u(0)] (\cos(n\theta + \theta_n) + 1) d\theta = \pi |c_n| R^n.$$

By the assumptions, $u(Re^{i\theta}) \leq A$, so the maximum principle gives $u(0) \leq A$, and

$$\begin{aligned} \int_0^{2\pi} [u(Re^{i\theta}) - u(0)] (\cos(n\theta + \theta_n) + 1) d\theta &\leq (A - u(0)) \int_0^{2\pi} (\cos(n\theta + \theta_n) + 1) d\theta \\ &= 2\pi(A - u(0)). \end{aligned}$$

Substituting, this becomes

$$\pi |c_n| R^n \leq 2\pi(A - u(0)).$$

Recall that $c_n = \frac{f^{(n)}(0)}{n!}$. Thus

$$|f^{(n)}(0)| \leq \frac{2n!}{R^n} (A - u(0)).$$

□

3 Entire Functions, by Andrew Lee

Definition 2 (Order). *The order of an entire function, f , is the infimum of all possible $\lambda > 0$ such that there exists constants A and B that satisfy*

$$|f(z)| \leq Ae^{B|z|^\lambda}$$

Note that an entire function, f , is of order less than λ if for all $\varepsilon > 0$, there exists A_ε and B_ε such that

$$|f(z)| \leq A_\varepsilon e^{B_\varepsilon |z|^{\lambda+\varepsilon}}$$

The next two propositions use the idea of order to bound the distribution of zeroes in an entire function.

Proposition 3.1. *Let f be an entire function of order λ with $f(0) = 1$. Then, for any $\varepsilon > 0$ there exists a constant, C_ε , that satisfies*

$$N_f(R) \leq C_\varepsilon R^{\lambda+\varepsilon}$$

Proof. We use the fact that $N_f(R)$ is nondecreasing and Jensen's Inequality to get

$$\begin{aligned}
N_f(R) &= \int_R^{eR} \frac{N_f(t)}{t} dt \\
&\leq \int_0^{eR} \frac{N_f(t)}{t} dt \\
&= \log M_f(eR) \\
&\leq \log A_\varepsilon e^{B_\varepsilon (eR)^{\lambda+\varepsilon}} \\
&\leq C_\varepsilon R^{\lambda+\varepsilon}
\end{aligned}$$

□

Proposition 3.2. *Let f be an entire function of order λ with $f(0) = 1$ and $a_1, a_2, \dots$ be the zeroes of f in non-decreasing order of norms. Then, for any $\varepsilon > 0$,*

$$\sum_{n=1}^{\infty} \frac{1}{|a_n|^{\lambda+\varepsilon}} < \infty$$

Proof. Separating the sum by powers of 2 then applying Proposition 2.1, we get

$$\begin{aligned}
\sum_{n=1}^{\infty} \frac{1}{|a_n|^{\lambda+\varepsilon}} &= \sum_{|a_n| < 1} \frac{1}{|a_n|^{\lambda+\varepsilon}} + \sum_{k=0}^{\infty} \sum_{2^k \leq |a_n| < 2^{k+1}} \frac{1}{|a_n|^{\lambda+\varepsilon}} \leq \infty \\
&\leq C + \sum_{k=0}^{\infty} \frac{N(2^{k+1})}{2^{(\lambda+\varepsilon)k}} \\
&\leq C + C_\varepsilon \sum_{k=0}^{\infty} \frac{2^{(\lambda+\varepsilon/2)(k+1)}}{2^{(\lambda+\varepsilon)k}} \\
&= C + C_\varepsilon \sum_{k=0}^{\infty} 2^{(\lambda+\varepsilon/2)-k\varepsilon/2} < \infty
\end{aligned}$$

□

Definition 3 (Convergence Index). *The convergence index of a sequence $a_1, a_2, \dots$ is the infimum of all ρ such that*

$$\sum_{n=1}^{\infty} \frac{1}{|a_n|^\rho} < \infty$$

Thus, Proposition 2.2 states that the convergence index of the zeroes of an entire function of order λ is less than λ .

4 Hadamard Factorization Theorem, by Toyesh Jayaswal

This section will be devoted to proving the following theorem to factorize certain complex valued functions. The theorem will be used later to factorize functions related to the Riemann zeta function

$\zeta(s)$ which can then be used to relate information about the distribution of prime numbers to the complex zeros of a function.

Theorem 4.1 (Hadamard Factorization Theorem). *Let $f(z)$ be a complex entire function of finite order λ and roots a_i . Then, $f(z)$ can be written as*

$$f(z) = e^{Q(z)} \prod_{n=1}^{\infty} E_p(z; a_n)$$

where $p = [\lambda]$, $E_p(w)$ is the canonical factor

$$E_p(z; a) = \left(1 - \frac{z}{a}\right) \exp\left(\sum_{k=1}^p \frac{z^k}{ka^k}\right),$$

and $Q(z)$ is a polynomial of degree at most p .

The theorem extends the property of polynomials to be factored based on their roots to entire functions of finite order. If a_i is the sequence of the zeros of a degree n polynomial function, that polynomial may be written as

$$f(z) = k \prod_{i=1}^n \left(1 - \frac{z}{a_i}\right).$$

The issue with extending the finite product to an infinite product is that the infinite product may diverge at points that are not zeros of the function. The solution is to add a term to each factor which will control the rate of growth for the trailing terms of the product.

Let a_n be the sequence of zeros of $f(z)$ ordered by non-decreasing absolute value. Since $\mathbb{C}$ is compact, a_n tends towards infinity since otherwise, a_n would have a limit point and $f(z)$ would be identically 0. Fix some R and choose N such that $|a_n| > 2R$ for $n > N$. Then, for $|z| < R$, $|z/a_n| < 1/2$ for $n > N$ so that $\log(1 - z/a_n)$ is analytic on $B(0, R)$. Expanding in terms of a power series,

$$\log\left(1 - \frac{z}{a_n}\right) = -\sum_{k=1}^{\infty} \frac{z^k}{ka_n^k}$$

Now, truncate the first $p = [\lambda]$ terms of the series and take the absolute value so that

$$\left|\log\left(1 - \frac{z}{a_n}\right) + \sum_{k=1}^p \frac{z^k}{ka_n^k}\right| = \left|\sum_{k=p+1}^{\infty} \frac{z^k}{ka_n^k}\right| \leq \frac{R^{p+1}}{a_n^{p+1}} \sum_{k=1}^{\infty} \frac{1}{2^k} < \frac{2R^{\lambda+\varepsilon}}{a_n^{\lambda+\varepsilon}}$$

with the last inequality holding because $|R/a_n| < 1/2$ and $\lambda + \varepsilon < p + 1$. Now,

$$\left|\sum_{n=N}^{\infty} \left(\log\left(1 - \frac{z}{a_n}\right) + \sum_{k=1}^p \frac{z^k}{ka_n^k}\right)\right| \leq \sum_{n=N}^{\infty} \frac{2R^{\lambda+\varepsilon}}{a_n^{\lambda+\varepsilon}} = 2R^{\lambda+\varepsilon} \sum_{n=N}^{\infty} \frac{1}{a_n^{\lambda+\varepsilon}}$$

which is $O(R^{\lambda+\varepsilon})$ because the convergence index of $a_n < \lambda$ and the infinite sum converges. Hence,

the series

$$\sum_{n=N}^{\infty} \left(\log \left(1 - \frac{z}{a_n} \right) + \sum_{k=1}^p \frac{z^k}{ka_n^k} \right)$$

is uniformly convergent for $|z| < R$ by the Weierstrass M test. For the $n \leq N$ terms, for $|z| = R$,

$$\begin{aligned} \left| \prod_{n=1}^N \left(1 - \frac{z}{a_n} \right) \exp \left(\sum_{k=1}^p \frac{z^k}{ka_n^k} \right) \right| &\leq \left(1 + \frac{R}{|a_1|} \right)^N \exp \left(C_p \sum_{n=1}^N \left| \frac{z}{a_n} \right|^p \right) \\ &\leq \exp(C_\varepsilon R^{\lambda+\varepsilon}) \exp \left(C_p R^p \sum_{n=1}^N \frac{|a_n|^{\lambda-p+\varepsilon}}{|a_n|^{\lambda+\varepsilon}} \right) \\ &\leq \exp \left(C_\varepsilon R^{\lambda+\varepsilon} + C_p R^{\lambda+\varepsilon} \sum_{n=1}^N \frac{1}{|a_n|^{\lambda+\varepsilon}} \right) \\ &\leq \exp(C_\varepsilon R^{\lambda+\varepsilon}) \end{aligned}$$

Where the first inequality is because a_n is ordered by magnitude and the first part of the second inequality follows from the bound of the number of zeros. Then, the product

$$\prod_{n=1}^{\infty} \left(1 - \frac{z}{a_n} \right) \exp \left(\sum_{k=1}^p \frac{z^k}{ka_n^k} \right) = \prod_{n=1}^{N-1} \times \prod_{n=N}^{\infty}$$

converges uniformly for $|z| < R$ since the first product is finite and the second is the exponential of the previous sum. Since R was arbitrary, the disk of convergence can be increased to the entire complex plane. The exponential factors do not contribute to the zeros of the product and hence, the product has the same zeros as $f(z)$. The product is bounded by $e^{C_\varepsilon R^{\lambda+\varepsilon}}$ so the order of the function is at most λ but because of the convergence index of the roots, the order must be at least λ , and hence the order of this new function is the same as that of $f(z)$.

The ratio of $f(z)$ with the infinite product is nonzero on the entire complex plane so that its logarithm is well defined everywhere. Let $Q(z)$ be the logarithm of this ratio. Then,

$$f(z) = e^{Q(z)} \prod_{n=1}^{\infty} \left(1 - \frac{z}{a_n} \right) \exp \left(\sum_{k=1}^{[\lambda]} \frac{z^k}{ka_n^k} \right)$$

The term

$$E_p(z; a) = \left(1 - \frac{z}{a} \right) \exp \left(\sum_{k=1}^p \frac{z^k}{ka^k} \right)$$

is known as the canonical factor.

All that is left to show is that $Q(z)$ is a polynomial of degree at most p . Consider the function

$$G_R(z) = f(z) \left(\prod_{|a_n| < R} E_p(z; a_n) \right)^{-1} = e^{Q(z)} \prod_{|a_n| \geq R} E_p(z; a_n)$$

Since the last product converges locally uniformly to 1 as $R \rightarrow \infty$, $\log(G_R)$ converges locally uniformly to Q . For $|a_n| < R$ and $|z| = 2R$, $|z/a_n| \geq 2$ so that

$$\begin{aligned} \prod_{|a_n| < R} E_p(a_n) &\geq \exp \left(-C_p \sum_{|a_n| < R} \left| \frac{z}{a_n} \right|^p \right) \\ &= \exp \left(-C_p R^p \sum_{|a_n| < R} \frac{|a_n|^{\lambda+\varepsilon-p}}{|a_n|^{\lambda+\varepsilon}} \right) \\ &\geq \exp \left(-C_p R^{\lambda+\varepsilon} \sum_{|a_n| < R} \frac{1}{|a_n|^{\lambda+\varepsilon}} \right) \\ &= \exp(-C_\varepsilon R^{\lambda+\varepsilon}) \end{aligned}$$

Where the first inequality is because $|1 - z/a_n| > 1$ and the exponential term is at least $e^2 > e^{-C_p 2^p}$. Hence, for $|z| = 2R$,

$$|G_R| \leq e^{(C_\varepsilon R^{\lambda+\varepsilon})} M_f(2R) \leq e^{(C_\varepsilon R^{\lambda+\varepsilon})}$$

by the definition of functions of order λ . By the maximum modulus principle,

$$\log |G_R| \leq C_\varepsilon R^{\lambda+\varepsilon}$$

for $|z| < R$. Hence, by the Borel-Caratheodory lemma,

$$|(\log G_R)^{(k)}(0)| \leq C_\varepsilon k! R^{\lambda+\varepsilon-k}$$

since $\operatorname{Re}(\log G_R) = \log |G_R|$. Then, for $k > \lambda \geq p$,

$$|Q^{(k)}(0)| = \lim_{R \rightarrow \infty} |(\log G_R)^{(k)}(0)| = 0$$

so that $Q(z)$ is a polynomial of degree at most p .

One important example of this theorem is that of the Γ function which we will encounter again later. The function is defined to be the analytic continuation of

$$\Gamma(z) = \int_0^\infty e^{-t} t^z \frac{dt}{t}.$$

Its reciprocal can be analytically continued to the whole complex plane to an entire function of order 1 with zeros $0, -1, -2, \dots$ and can be written (as shown in chapter 7 of [2]) as

$$\frac{1}{\Gamma(z)} = z e^{\gamma z} \prod_{n=1}^{\infty} \left(1 + \frac{z}{n} \right) e^{-z/n}$$

with γ is chosen so that $\Gamma(1) = 1$ and is known as the Euler-Mascheroni constant. This expansion is

in the form prescribed by the Hadamard factorization theorem. This then implies that

$$\Gamma(z) = \frac{e^{-\gamma z}}{z} \prod_{n=1}^{\infty} \left(1 + \frac{z}{n}\right)^{-1} e^{z/n} \quad (1)$$

5 Chebyshev's Functions, by Ishita Goluguri

In this section, we define two functions, $\psi(x)$ and $\vartheta(x)$, that are related to the prime counting function.

Definition 4. We define the Von-Mangoldt function Λ as the following arithmetic function:

$$\Lambda(n) = \begin{cases} \log p & \text{if } n = p^m \text{ for some prime } p \text{ and some } m \geq 1 \\ 0 & \text{otherwise} \end{cases}$$

Based on this, we define two Chebyshev functions ψ and ϑ for positive real variable x as

$$\psi(x) = \sum_{n \leq x} \Lambda(n), \quad \vartheta(x) = \sum_{p \leq x} \log p.$$

Note that

$$\psi(x) = \sum_{n \leq x} \Lambda(n) = \sum_{m=1}^{\infty} \sum_{p^m \leq x} \Lambda(p^m) = \sum_{m=1}^{\infty} \sum_{p \leq x^{1/m}} \log p = \sum_{m=1}^{\infty} \vartheta(x^{1/m})$$

Relationships between these functions and $\pi(x)$ can be deduced using the Abel's identity which gives an approximation for certain sums to understand their asymptotic behavior. Essentially, the sum can be approximated by values at the end points with the error given by an integral. We cite it from [1].

Theorem 5.1 (Abel's Identity). *Let $a(n)$ be an arithmetical function and $f(x)$ be a function differentiable on the interval $[y, x]$. Define*

$$A(x) = \sum_{n=1}^x a(n)$$

for $x > 1$ and $A(x) = 0$ for $0 < x < 1$. Then,

$$\sum_{y < n \leq x} a(n)f(n) = A(x)f(x) - A(y)f(y) - \int_y^x f'(t)A(t)dt$$

Next, we prove a theorem that will be used with $a(n)$ in the above as the characteristic function of primes to show relations between the Chebyshev and Prime Counting functions. Using Abel's identity, some relations can be found between $\vartheta(x)$ and $\pi(x)$ from which the asymptotic behavior of $\pi(x)$ and the Chebyshev functions can be seen to be related.

Theorem 5.2.

$$\vartheta(x) = \pi(x) \log x + \int_2^x \frac{\pi(t)}{t} dt$$

$$\pi(x) = \frac{\vartheta(x)}{\log x} + \int_2^x \frac{\vartheta(t)}{t \log^2 t} dt$$

Proof. Let $a(n)$ be the characteristic function of primes so that $a(n) = 1$ if n is prime and 0 otherwise. Then,

$$\pi(x) = \sum_{p \leq x} 1 = \sum_{n=2}^x a(n)$$

and

$$\vartheta(x) = \sum_{p \leq x} \log(p) = \sum_{n=2}^x a(n) \log(n)$$

By the Abel summation formula,

$$\vartheta(x) = \log x \sum_{n=2}^x a(n) - \log(1)a(1) - \int_1^x \frac{1}{t} \sum_{n=2}^t a(n) dt = \pi(x) \log x - \int_1^x \frac{\pi(t)}{t} dt$$

which proves the first claim since $\pi(x) = 0$ for $x < 2$. Similarly, let $b(n) = a(n) \log(n)$ so that

$$\pi(x) = \sum_{n > 3/2}^x \frac{b(n)}{\log(n)}$$

and

$$\vartheta(x) = \sum_{n > 3/2}^t b(n)$$

By the Abel summation formula,

$$\begin{aligned} \pi(x) &= \frac{1}{\log x} \sum_{n=2}^x b(n) - \frac{b(3/2)}{\log(3/2)} - \int_{3/2}^x \frac{1}{t \log^2(x)} \sum_{n > 3/2}^t b(n) dt \\ &= \frac{\vartheta(x)}{\log x} - \frac{b(3/2)}{\log(3/2)} - \int_{3/2}^x \frac{\vartheta(t)}{t \log^2(t)} dt. \end{aligned}$$

which proves the second claim since $\vartheta(x)$ and $b(x)$ are both 0 for $x < 2$. $\square$

Studying the prime counting function $\pi(x)$ is equivalent to studying $\vartheta(x)$ and $\psi(x)$. In fact, the prime number theorem,

$$\lim_{x \rightarrow \infty} \frac{\pi(x) \log x}{x} = 1$$

is equivalent to

$$\sum_{n \leq x} \Lambda(n) \sim x$$

as $x \rightarrow \infty$. By definition, this is equivalent to $\vartheta(x) \sim x$

By definition, $\Lambda(n) = 0$ unless n is the power of a prime.

$$\psi(x) = \sum_{n \leq x} \Lambda(n) = \sum_{m=1}^{\infty} \sum_p \Lambda(p^m) = \sum_{m=1}^{\infty} \sum_{p \leq x^{1/m}} \log p.$$

Thus,

$$\psi(x) = \sum_{m \leq \log_2 x} \sum_{p \leq x^{1/m}} \log p.$$

Using the definition of ϑ ,

$$\psi(x) = \sum_{m \leq \log_2 x} \vartheta(x^{1/m}).$$

Theorem 5.3. *This theorem relates $\psi(x)/x$ and $\vartheta(x)/x$. For $x > 0$,*

$$0 \leq \frac{\psi(x)}{x} \leq \frac{\vartheta(x)}{x} \leq \frac{(\log x)^2}{2\sqrt{x} \log 2}.$$

This inequality implies that

$$\lim_{x \rightarrow \infty} \left(\frac{\psi(x)}{x} - \frac{\vartheta(x)}{x} \right) = 0.$$

Proof. Using the relation between ϑ and ψ ,

$$0 \leq \psi(x) - \vartheta(x) = \sum_{2 \leq m \leq \log_2(x)} \vartheta(x^{\frac{1}{m}}).$$

From the definition of $\vartheta(x)$, we have $\vartheta(x) \leq x \log x$. Combining all the statements,

$$0 \leq \psi(x) - \vartheta(x) \leq \sum_{2 \leq m \leq \log_2 x} x^{\frac{1}{m}} \log(x^{\frac{1}{m}}) \leq \log_2 x \sqrt{x} \log \sqrt{x} = \frac{\sqrt{x}(\log x)^2}{2 \log 2}.$$

Dividing by x , we get the desired statement:

$$0 \leq \frac{\psi(x)}{x} \leq \frac{\vartheta(x)}{x} \leq \frac{(\log x)^2}{2\sqrt{x} \log 2}.$$

□

Theorem 5.4. *The following statements are all logically equivalent.*

$$\lim_{x \rightarrow \infty} \frac{\pi(x) \log x}{x} = 1$$

$$\lim_{x \rightarrow \infty} \frac{\vartheta(x)}{x} = 1$$

$$\lim_{x \rightarrow \infty} \frac{\psi(x)}{x} = 1$$

Proof. Using the statements of theorem 4.1,

$$\frac{\vartheta(x)}{x} = \frac{\pi(x) \log x}{x} - \frac{1}{x} \int_2^x \frac{\pi(t)}{t} dt.$$

and

$$\frac{\pi(x) \log x}{x} = \frac{\vartheta(x)}{x} + \frac{\log x}{x} \int_2^x \frac{\vartheta(t) dt}{t \log^2 t}$$

To show the relation between the first two statements, it suffices to show that $\lim_{x \rightarrow \infty} \frac{1}{x} \int_2^x \frac{\pi(t)}{t} dt = 0$. The statement $\lim_{x \rightarrow \infty} \frac{\pi(x) \log x}{x} = 1$ implies $\frac{\pi(t)}{t} = O\left(\frac{1}{\log t}\right)$ for $t \geq 2$. Thus,

$$\frac{1}{x} \int_2^x \frac{\pi(t)}{t} dt = O\left(\frac{1}{x} \int_2^x \frac{dt}{\log t}\right).$$

By L'Hospital's rule, the right-hand-side has limit zero as $x \rightarrow \infty$. This proves that $\lim_{x \rightarrow \infty} \frac{1}{x} \int_2^x \frac{\pi(t)}{t} dt = 0$.

To show the opposite direction between these two statements, it suffices to prove that

$$\lim_{x \rightarrow \infty} \frac{\vartheta(x)}{x} = 1 \implies \lim_{x \rightarrow \infty} \frac{\pi(x) \log x}{x} = 1.$$

We've proven that

$$\lim_{x \rightarrow \infty} \frac{\pi(x) \log x}{x} = 1 \implies \lim_{x \rightarrow \infty} \frac{\vartheta(x)}{x} = 1$$

and thus that $\vartheta(x) = O(x)$. Thus,

$$\frac{\log x}{x} \int_2^x \frac{\vartheta(t) dt}{t \log^2 t} = O\left(\frac{\log x}{x} \int_2^x \frac{dt}{\log^2 t}\right).$$

Again by L'Hospital's rule, the right-hand-side has limit zero as $x \rightarrow \infty$. Thus, this proves that the second statement implies the first. The connection between the second and third statements has already been proven in theorem 4.3, so all three of these statements are equivalent. $\square$

6 Perron Formula, by Andrew Lee

The Perron Formula acts as a filter to isolate the first finitely many terms from a Dirichlet series. The formula will use the following lemma:

Lemma 6.1. For $c > 0$ and having $\int_{c-i\infty}^{c+i\infty}$ to mean $\lim_{T \rightarrow \infty} \int_{c-iT}^{c+iT}$,

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{a^z}{z} dz = \begin{cases} 1 & a > 1 \\ \frac{1}{2} & a = 1 \\ 0 & 0 < a < 1 \end{cases}$$

Proof. For $0 < a < 1$, consider the contour $\gamma : [c-iT, c+iT, b+iT, b-iT, c-iT]$ for some $b > c$ so that it is a rectangle to the right of the line of interest. Since $\frac{a^z}{z}$ is analytic on the interior of γ , $\int_{\gamma} \frac{a^z}{z} dz = 0$ so that

$$\int_{c-iT}^{c+iT} = -\left(\int_{c+iT}^{b+iT} + \int_{b+iT}^{b-iT} + \int_{b-iT}^{c-iT}\right)$$

and

$$\begin{aligned} \left| \int_{c-iT}^{c+iT} \frac{a^x}{T} dx \right| &\leq \int_c^b \frac{a^x}{T} dx + \int_{-T}^T \frac{a^b}{b} dy + \int_c^b \frac{a^x}{T} dx \\ &\leq \frac{2}{T} \int_c^\infty a^x dx + \frac{2Ta^b}{b} \\ &= \frac{2}{T} \frac{-a^c}{\log a} + \frac{2Ta^b}{b} \end{aligned}$$

As $b \rightarrow \infty$, $a^b \rightarrow 0$ since $a < 1$ and as $T \rightarrow \infty$, the first term also vanishes which proves the first claim.

For $a > 1$, consider the contour $\gamma : [c - iT, c + iT, -b + iT, -b - iT, c - iT]$ for some $b > 0$ so that it is a rectangle to the left of the line of interest containing 0. In this region $\frac{a^z}{z}$ has only one pole at 0 with residue 1 so that

$$\int_{c-iT}^{c+iT} + \int_{c+iT}^{-b+iT} + \int_{-b+iT}^{-b-iT} + \int_{-b-iT}^{c-iT} = 2\pi i$$

and

$$\frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{a^z}{z} dz - 1 = \frac{-1}{2\pi i} \left(\int_{c+iT}^{-b+iT} + \int_{-b+iT}^{-b-iT} + \int_{-b-iT}^{c-iT} \right)$$

Now, the integrals on the right can be estimated as

$$\begin{aligned} \left| - \int_{c+iT}^{-b+iT} \frac{a^z}{z} dz \right| &\leq \int_{-b}^c \frac{a^x}{T} dx \leq \int_{-\infty}^c \frac{a^x}{T} dx = \frac{a^c}{T \log a} \\ \left| - \int_{-b+iT}^{-b-iT} \frac{a^z}{z} dz \right| &\leq \int_{-T}^T \frac{a^{-b}}{b} dy = \frac{2Ta^{-b}}{b} \\ \left| - \int_{-b-iT}^{c-iT} \frac{a^z}{z} dz \right| &\leq \int_{-\infty}^c \frac{a^x}{T} dx = \frac{a^c}{T \log a} \end{aligned}$$

As $b \rightarrow \infty$, the second integral vanishes and as $T \rightarrow \infty$ the other ones do as well which proves the last claim.

For $a = 1$.

$$\int_{c-iT}^{c+iT} \frac{1}{z} dt = \int_{-T}^T \frac{i}{c + iy} dy = \int_{-T}^T \frac{ic}{c^2 + y^2} dy + \int_{-T}^T \frac{y}{c^2 + y^2} dy$$

and the second integral vanishes since the integrand is an odd function. Now,

$$\frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{1}{z} dt = \frac{1}{\pi} \int_0^T \frac{c}{c^2 + y^2} dy = \frac{1}{\pi} \arctan\left(\frac{T}{c}\right) = \frac{1}{2} - \frac{1}{\pi} \arctan\left(\frac{c}{T}\right)$$

and the $\arctan(c/T)$ term vanishes as $T \rightarrow \infty$. □

Theorem 6.2 (Perron Formula). *Let $F(s) = \sum_{n=1}^\infty f(n)/n^s$ be absolutely convergent for $\sigma > \sigma_a$. Then for arbitrary $c, x > 0$, if $\sigma > \sigma_a - c$,*

$$\frac{1}{2\pi i} \int_{c-\infty i}^{c+\infty i} F(s+z) \frac{x^z}{z} dz = \sum_{n \leq x}^* \frac{f(n)}{n^s}$$

where $\sum^*$ means that the last term is halved when x is an integer.

Proof. We note that $F(s+z)$ is absolutely and uniformly convergent on the half plane $\sigma > \sigma_a - c$ since the real part of z is c . Thus,

$$\begin{aligned} \int_{c-iT}^{c+iT} F(s+z) \frac{x^z}{z} dz &= \int_{c-iT}^{c+iT} \sum_{n=1}^{\infty} \frac{f(n)}{n^{s+z}} \frac{x^z}{z} dz \\ &= \sum_{n=1}^{\infty} \frac{f(n)}{n^s} \int_{c-iT}^{c+iT} \left(\frac{x}{n}\right)^z \frac{dz}{z} \\ &= \sum_{n < x} \frac{f(n)}{n^s} \int_{c-iT}^{c+iT} \left(\frac{x}{n}\right)^z \frac{dz}{z} + \sum_{n > x} \frac{f(n)}{n^s} \int_{c-iT}^{c+iT} \left(\frac{x}{n}\right)^z \frac{dz}{z} +' \frac{f(x)}{x^s} \int_{c-iT}^{c+iT} \frac{dz}{z} \end{aligned}$$

where $+'$ denotes that the last term is only present if x is an integer.

By the previous lemma, as T tends to infinity, each integral in the $\sum_{n < x}$ term simplifies to $2\pi i$. Also, if the last term is present, it simplifies to $\frac{f(x)}{x^s} \pi i$. All that is left to prove is that

$$\lim_{T \rightarrow \infty} \sum_{n > x} \frac{f(n)}{n^s} \int_{c-iT}^{c+iT} \left(\frac{x}{n}\right)^z \frac{dz}{z} = 0.$$

Note that since $n \geq [x] + 1$, $n/x \geq ([x] + 1)/x$. Thus, using the previous lemma again, we have

$$\begin{aligned} \left| \sum_{n > x} \frac{f(n)}{n^s} \int_{c-iT}^{c+iT} \left(\frac{x}{n}\right)^z \frac{dz}{z} \right| &\leq \sum_{n > x} \frac{|f(n)|}{n^\sigma} \frac{2}{t} \left(\frac{x}{n}\right)^c \frac{1}{\log\left(\frac{1}{[x]+1}\right)} \\ &= \frac{2}{T} \frac{x^c}{\log\left(\frac{1}{[x]+1}\right)} \sum_{n > x} \frac{|f(n)|}{n^{\sigma+c}} \end{aligned}$$

which tends to 0 as T approaches infinity. This concludes the proof. $\square$

We are going to use the Perron formula to deduce an integral representation for $\psi(x)$.

Definition 5 (Reimann ζ Function).

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

Theorem 6.3. The ζ function for $s > 1$ can be written as

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}}$$

where the product is taken over all prime numbers.

Proof. Expanding each term in the product as a geometric sum,

$$\begin{aligned}\prod_p \frac{1}{1-p^{-s}} &= \prod_p \left(\sum_{k=0}^{\infty} \frac{1}{p^{ks}} \right) \\ &= \left(1 + \frac{1}{2^s} + \frac{1}{4^s} + \dots \right) \left(1 + \frac{1}{3^s} + \frac{1}{9^s} + \dots \right) \left(1 + \frac{1}{5^s} + \frac{1}{25^s} + \dots \right) \dots\end{aligned}$$

Expanding out the products, each integer n appears exactly once. This is just a restatement of the fundamental theorem of arithmetic that each number has a unique prime factorization. For example, $1/140^s$ would take $1/4^s$ from the first factor, $1/5^s$ from the third, $1/7^s$ from the fourth, and 1 from all other factors. Hence,

$$\prod_p \left(\sum_{k=0}^{\infty} \frac{1}{p^{ks}} \right) = 1 + \frac{1}{2^s} + \frac{1}{3^s} + \dots = \zeta(s)$$

for $s > 1$. □

This relationship with only the prime numbers on one side and all the natural numbers on the other allows us to be able to derive information about the prime numbers from the ζ function.

Theorem 6.4. For $x > 0$, not equal to an integer,

$$\int_0^x \frac{\psi(y)}{y} dy = \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} -\frac{\zeta'}{\zeta}(s) \frac{x^s}{s^2} ds$$

Proof. Recall that

$$\psi(x) = \sum_{n \leq x} \Lambda(n)$$

Also, from the Euler representation of ζ , we have

$$\log \zeta(s) = - \sum_p \log(1 - p^{-s}), \quad \sigma > 1.$$

Differentiating,

$$\frac{\zeta'}{\zeta}(s) = - \sum_p \log p \cdot \frac{p^{-s}}{1 - p^{-s}} = - \sum_p \log p \sum_{m=1}^{\infty} p^{-ms} = - \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}, \quad \sigma > 1.$$

Then, applying the Perron Formula,

$$\psi(x) = \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} -\frac{\zeta'}{\zeta}(s) \frac{x^s}{s} ds.$$

Dividing both sides by x and then integrate, we obtain the desired result. □

7 Reimann ζ Function, by Ishita Goluguri

To recap, a function is holomorphic if it's complex-differentiable in each of its neighborhoods. Because the series converges locally uniformly for $\sigma > 1$, $\zeta(s)$ is holomorphic on the right half-plane $\sigma > 1$ and never assumes the value zero on that half-plane. Using this property along with the definitions of two functions- the Gamma function and Poisson summation function- we can define a useful relation between the three.

7.1 The Functional Equation

By a simple calculation,

$$(\sqrt{\pi}n)^{-s}\Gamma\left(\frac{s}{2}\right) = \int_0^\infty x^{s/2-1}e^{-n^2\pi x}dx.$$

Summing over n gives

$$\begin{aligned}\frac{\Gamma\left(\frac{s}{2}\right)\zeta(s)}{\pi^{s/2}} &= \sum_{n=1}^\infty \int_0^\infty x^{s/2-1}e^{-n^2\pi x}dx \\ &= \int_0^\infty x^{s/2-1}\Psi(x)dx \\ &= \int_0^1 x^{s/2-1}\Psi(x)dx + \int_1^\infty x^{s/2-1}\Psi(x)dx\end{aligned}$$

where $\Psi(x)$ is defined as

$$\Psi(x) = \sum_{n=1}^\infty e^{-\pi n^2 x}.$$

Using the Poisson summing formula (see e.g. Chapter 18 of Zorich [11]), we have

$$2\Psi(x) + 1 = \frac{1}{\sqrt{x}} \left(2\Psi\left(\frac{1}{x}\right) + 1 \right). \quad (2)$$

Substituting this into the integral from 0 to 1,

$$\begin{aligned}\frac{\Gamma\left(\frac{s}{2}\right)\zeta(s)}{\pi^{s/2}} &= \int_0^1 x^{s/2-1} \left(\frac{1}{\sqrt{x}}\Psi\left(\frac{1}{x}\right) + \frac{1}{2\sqrt{x}} - \frac{1}{2} \right) dx + \int_1^\infty x^{s/2-1}\Psi(x)dx \\ &= \frac{1}{s(s-1)} + \int_0^1 x^{s/2-3/2}\Psi\left(\frac{1}{x}\right) + \int_1^\infty x^{s/2-1}\Psi(x)dx.\end{aligned}$$

Next, we wish to make this an integral from 1 to infinity. To do this, we change the variable $x \rightarrow \frac{1}{x}$

$$\frac{\Gamma\left(\frac{s}{2}\right)\zeta(s)}{\pi^{s/2}} = \frac{1}{s(s-1)} + \int_1^\infty \left(x^{-s/2-1/2} + x^{s/2-1} \right) \Psi(x)dx$$

Define this function to be $\xi(s)$:

$$\xi(s) = \frac{\Gamma\left(\frac{s}{2}\right)\zeta(s)}{\pi^{s/2}}. \quad (3)$$

Substituting in $s = 1 - s$, we see that $\xi(s)$ satisfies the functional equation

$$\xi(s) = \xi(1 - s).$$

Thus, we've found an analytic continuation of $\zeta(s)$ to the entire complex plane: it is holomorphic except for the first order pole $s = 1$, where its residue is 1.

7.2 The Factorization of $\zeta(s)$ and Explicit Formula for Prime Numbers

We find that $\xi(s)$ is asymptotically related to $\Gamma(s)$. In fact, since

$$e^{-\pi x} < \Psi(x) < \sum_{n=1}^{\infty} e^{-\pi nx} = \frac{e^{-\pi x}}{1 - e^{-\pi x}} < 2e^{-\pi x}, \quad x > 1,$$

we can bound, for $s \geq 1$ real,

$$\left| \int_1^{\infty} \left(x^{-s/2-1/2} + x^{s/2-1} \right) \Psi(x) dx \right| \leq 4 \int_1^{\infty} x^{s/2} e^{-\pi x} dx \leq C \Gamma\left(\frac{s}{2}\right) = O(e^{s \log s}),$$

and

$$\int_1^{\infty} \left(x^{-s/2-1/2} + x^{s/2-1} \right) \Psi(x) dx \geq \int_1^{\infty} x^{s/2-1} e^{-\pi x} dx \geq c \Gamma\left(\frac{s}{2}\right).$$

Thus, $s(s-1)\xi(s)$ is holomorphic over all points of the complex plane and is a first order entire function. When $s \rightarrow +\infty$,

$$\xi(s) \sim \Gamma\left(\frac{s}{2}\right).$$

Let the set of zeroes of $\xi(s)$ be $\{\rho\}$. If $\{\rho\}$ was finite, then by the Hadamard factorization theorem, we must have $\xi(s) = P(s) \cdot e^{as}$, where $P(s)$ is a polynomial and a is a constant, whence we have $|\xi(s)| = O(e^{2|a||s|})$. However, we know from Stirling's formula that

$$|\xi(s)| \sim \Gamma\left(\frac{s}{2}\right) \sim e^{s \log s}, \quad s \rightarrow +\infty,$$

so $\{\rho\}$ must be infinite.

By (3) and the Euler product formula, $\xi(s)$ has no zeroes with $\operatorname{Re}(s) > 1$. Using the functional equation $\xi(s) = \xi(1-s)$, it also has no zeroes with $\operatorname{Re}(s) < 0$. Thus, all zeroes of $\xi(s)$ are within $0 \leq \sigma \leq 1$.

Using the functional equation $\xi(s) = \xi(1-s)$ and $\bar{\xi}(s) = \xi(\bar{s})$, we see that the distribution of zeroes is symmetric with respect to the $\sigma = 1/2$ line and the real axis. If we let the number of zeroes with $|\operatorname{Im}(\rho)| \leq T$, then $N(T)$, then by Jensen's inequality, $N(T) \leq C_{\epsilon} T^{1+\epsilon}$ for any $\epsilon > 0$. Using the Hadamard Factorization Theorem, we obtain

$$\xi(s) = \frac{e^{as+b}}{s(1-s)} \prod_{\rho} \left(1 - \frac{s}{\rho} \right) e^{s/\rho}.$$

Thus, using formula (1), we find that

$$\zeta(s) = \frac{e^{A+Ds}}{s-1} \prod_{\rho} \left(1 - \frac{s}{\rho}\right) e^{s/\rho} \prod_{n=1}^{\infty} \left(1 + \frac{s}{2n}\right) e^{-s/(2n)}$$

where A and D are constants that aren't important for us.

We should also remember that $\zeta(s)$ has a first order pole at $s = 1$. Thus,

$$\frac{\zeta'}{\zeta}(s) = D - \frac{1}{s-1} + \sum_{\rho} \left(\frac{1}{s-\rho} + \frac{1}{\rho}\right) + \sum_{n=1}^{\infty} \left(\frac{1}{s+2n} - \frac{1}{2n}\right).$$

Calculating for $x > 1$, e.g.

$$\frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} \frac{x^s}{(s-1)s^2} ds = \lim_{T \rightarrow \infty} \frac{1}{2\pi i} \int_{2-iT}^{2+iT} \frac{x^s}{(s-1)s^2} ds.$$

We complete the line $[2-iT, 2+iT]$ with the left half circle centered at $s = 0$ with radius $\sqrt{T^2 + 4}$, and call this contour γ . The residues of $\frac{x^s}{(s-1)s^2}$ are x at $s = 1$ and $-\log x - 1$ at $s = 0$. By the residue theorem,

$$\frac{1}{2\pi i} \int_{\gamma} \frac{x^s}{s^2(s-1)} ds = x - \log x - 1.$$

The value of the integral is far smaller on the arc than on the lines, so we find that this is the limit we seek.

We may calculate the other integrals in a similar manner, and finally get the explicit formula for primes:

$$\int_0^x \frac{\psi(y)}{y} dy = x - (D+2) \log x - \sum_{\rho} \frac{x^{\rho}}{\rho^2} + \left(\frac{\pi^2}{24} + \sum_{\rho} \frac{1}{\rho^2}\right) - \sum_{n=1}^{\infty} \frac{x^{-2n}}{4n^2}.$$

8 The Prime Number Theorem, by Andrew Lee

In this section, we prove the prime number theorem, $\pi(x) \simeq x/\log x$. In Section 4, we proved that this is equivalent to $\int_0^x \frac{\psi(y)}{y} dy \simeq x$. Then, we want to prove that the x term dominates in the explicit formula for $\int_0^x \frac{\psi(y)}{y} dy$ described in the last section.

8.1 The Prime Number Theorem

Let $\rho = \beta + i\gamma$. Looking at the $\sum_{\rho} \frac{x^{\rho}}{\rho^2}$ term, we expect that $\beta < 1$. This was proved independently by Hadamard and de la vallée Poussin in 1896.

We follow de la vallée Poussin's method. Let $s = \sigma + it$. Then, taking the logarithm of the Euler product of $\zeta(s)$, we get

$$\begin{aligned}
\log |\zeta(\sigma + it)| &= -\operatorname{Re} \sum_p \log(1 - p^{-(\sigma + it)}) \\
&= \operatorname{Re} \sum_p \sum_{n=1}^{\infty} \frac{1}{np^{n(\sigma + it)}} \\
&= \sum_p \sum_{n=1}^{\infty} \frac{1}{np^{n\sigma}} \cos(nt \log p).
\end{aligned}$$

So

$$\begin{aligned}
&3 \log |\zeta(\sigma)| + 4 \log |\zeta(\sigma + it)| + \log |\zeta(\sigma + 2it)| \\
&= \sum_p \sum_{n=1}^{\infty} \frac{1}{np^{n\sigma}} (3 + 4 \cos(nt \log p) + \cos(2nt \log p)) \\
&= \sum_p \sum_{n=1}^{\infty} \frac{1}{np^{n\sigma}} (3 + 4 \cos(nt \log p) + 2 \cos^2(nt \log p) - 1) \\
&= \sum_p \sum_{n=1}^{\infty} \frac{1}{np^{n\sigma}} 2(\cos(nt \log p) + 1)^2 \geq 0,
\end{aligned}$$

Thus,

$$|\zeta(\sigma)|^3 |\zeta(\sigma + it)|^4 |\zeta(\sigma + 2it)| \geq 1, \sigma > 1.$$

If we assume $1 + it$ is a zero of $\zeta(s)$ of order $m \geq 1$, and let $\sigma \downarrow 1$ in the above inequality, we have

$$|\zeta(\sigma)|^3 \simeq \frac{1}{(\sigma - 1)^3}, |\zeta(\sigma + it)|^4 \simeq (\sigma - 1)^{4m},$$

and $|\zeta(\sigma + 2it)|$ is bounded. So

$$|\zeta(\sigma)|^3 |\zeta(\sigma + it)|^4 |\zeta(\sigma + 2it)| \leq C(\sigma - 1)^{4m-3},$$

which approaches 0, a contradiction.

Thus, all ρ lie in the strip $0 < \operatorname{Re}(\rho) < 1$. So there is a continuous non-increasing function $h : [0, \infty) \rightarrow (0, 1)$, such that $\zeta(s)$ is zero-free in the region

$$\sigma < h(t).$$

Then, estimating the summation over ρ in the explicit formula, we get

$$\begin{aligned}
\left| \sum_{\rho} \frac{x^{\rho}}{\rho^2} \right| &\leq \sum_{\rho: |\gamma| < T} + \sum_{\rho: |\gamma| \geq T} \frac{x^{\beta}}{|\rho|^2} \\
&\leq x^{h(T)} + x \sum_{k=0}^{\infty} \sum_{2^k T < |\gamma| \leq 2^{k+1} T} \frac{1}{|\rho|^2} \\
&\leq x^{h(T)} + x \sum_{k=0}^{\infty} \frac{N(2^{k+1} T)}{2^{2k} T^2} \\
&<<_{\varepsilon} x^{h(T)} + x T^{-(1-\varepsilon)}.
\end{aligned}$$

Suppose $x > 2$. Then, let $T = T(x)$ be the solution of

$$1 - h(T) = (1 - \varepsilon) \log T / \log x.$$

There always exists a solution since $1 - h(T)$ is less than 1 and $(1 - \varepsilon) \log T / \log x$ ranges from 0 to infinity. Note that as x approaches infinity, $T(x)$ approaches infinity as well. Rearranging, this value of T satisfies

$$x^{h(T)} = x T^{-(1-\varepsilon)}.$$

Then,

$$\left| \sum_{\rho} \frac{x^{\rho}}{\rho^2} \right| \leq x^{h(T)} + x T^{-(1-\varepsilon)} \leq C_{\varepsilon} x T^{-(1-\varepsilon)} = o(x)$$

Thus,

$$\int_0^x \frac{\psi(y)}{y} dy = x + o(x),$$

which gives us the prime number theorem. This shows that the more we increase the bounds on the zero-free region, the better our precision of our estimate of the error will be.

8.2 Estimate on the Error Term

Theorem 8.1. *There is a constant $A > 0$ such that $\zeta(s)$ has no zero in the region*

$$\sigma < 1 - \frac{A}{\log(2+t)}, t \geq 0.$$

Proof. We start from the expansion of ζ'/ζ . The sum over natural numbers can be estimated and we find that for $\text{Re}(s) > 0$,

$$\frac{\zeta'}{\zeta}(s) = -\frac{1}{s-1} + \sum_{\rho} \left(\frac{1}{s-\rho} + \frac{1}{\rho} \right) + O(\log(|s|+2)). \quad (4)$$

We also know for $\sigma > 1$,

$$\begin{aligned} & -3\operatorname{Re}\frac{\zeta'}{\zeta}(\sigma) - 4\operatorname{Re}\frac{\zeta'}{\zeta}(\sigma + it) - \operatorname{Re}\frac{\zeta'}{\zeta}(\sigma + 2it) \\ & = \sum_p \sum_{n=1}^{\infty} \frac{1}{p^{n\sigma}} (3 + 4\cos(nt \log p) + \cos(2nt \log p)) \geq 0. \end{aligned} \quad (5)$$

Let $\sigma \downarrow 1$. Then,

$$-\operatorname{Re}\frac{\zeta'}{\zeta}(\sigma) < \frac{2}{\sigma - 1}. \quad (6)$$

Now suppose $t > 2$. Taking real part in (1), we find

$$-\operatorname{Re}\frac{\zeta'}{\zeta}(\sigma + it) = -\sum_{\rho} \left(\frac{\sigma - \beta}{(\sigma - \beta)^2 + (t - \gamma)^2} + \frac{\beta}{\beta^2 + \gamma^2} \right) + O(\log t). \quad (7)$$

Since $\sigma > 1$, $0 < \beta < 1$, every term in the summation in (4) is positive. Thus,

$$-\operatorname{Re}\frac{\zeta'}{\zeta}(\sigma + it) < O(\log t). \quad (8)$$

If $\rho_0 = \beta_0 + i\gamma_0$ is a zero with $\gamma_0 > 2$, then we set $t = \gamma_0$ in (4), to obtain

$$-\operatorname{Re}\frac{\zeta'}{\zeta}(\sigma + i\gamma_0) < -\frac{1}{\sigma - \beta_0} + O(\log \gamma_0). \quad (9)$$

Substituting (3),(5),(6) back to (2), we see that

$$\frac{6}{\sigma - 1} - \frac{4}{\sigma - \beta_0} \geq -A_1 \log \gamma_0.$$

We can then solve β_0 and find

$$1 - \beta_0 \geq \frac{1 - (\sigma - 1)A_1 \log \gamma_0}{6(\sigma - 1)^{-1} + A_1 \log \gamma_0}.$$

We set $\sigma = 1 + A_1/(2 \log \gamma_0)$ and finally get

$$1 - \beta_0 \geq \frac{A}{\log \gamma_0}$$

if $\gamma_0 > 2$. Since there are only finitely many zeroes with $|\gamma| \leq 2$, we obtain the desired result. $\square$

Substituting this back to the explicit formula, we find the following by taking $T = e^{\sqrt{\log x}}$: for $x > 1$,

$$\left| \int_0^x \frac{\psi(y)}{y} dy - x \right| < bxe^{-c_1 \sqrt{\log x}}.$$

We set $\delta = \delta(x) = be^{-c_1\sqrt{\log x}}$, so the above formula reduces to

$$(1 - \delta)x < \int_0^x \frac{\psi(y)}{y} dy < (1 + \delta)x.$$

Since $\psi(x)$ is increasing, $\delta(x)$ is decreasing, we find

$$\frac{(1 + \sqrt{\delta})x - x}{(1 + \sqrt{\delta})x} \psi(x) \leq \int_x^{(1+\sqrt{\delta})x} \frac{\psi(y)}{y} dy < (1 + \delta)(1 + \sqrt{\delta})x - (1 - \delta)x$$

$$\frac{\sqrt{\delta}}{1 + \sqrt{\delta}} \psi(x) < (2\delta + \sqrt{\delta} + \delta\sqrt{\delta})x,$$

so

$$\psi(x) - x < (3\sqrt{\delta} + 3\delta + \delta\sqrt{\delta})x \leq (3b + 4)xe^{-c\sqrt{\log x}},$$

Similarly

$$\psi(x) > (-3\sqrt{\delta} + \delta + \delta\sqrt{\delta})x \geq -3bxe^{-c\sqrt{\log x}}.$$

Then,

$$\psi(x) = x + O(xe^{-c\sqrt{\log x}}).$$

Substituting this result into $\vartheta(x) = \psi(x) + O(\sqrt{x} \log x)$, we get

$$\vartheta(x) = x + O(xe^{-c\sqrt{\log x}}).$$

Substituting to the formula

$$\pi(x) = \frac{\vartheta(x)}{\log x} + \int_2^x \frac{\vartheta(y)}{y \log^2 y} dy,$$

we find

$$\begin{aligned} \pi(x) &= \frac{x}{\log x} + \int_2^x \frac{1}{\log^2 y} dy \\ &\quad + O(xe^{-c\sqrt{\log x}}) + \int_2^x \frac{O(ye^{-c\sqrt{\log y}})}{y \log^2 y} dy \\ &= \text{Li}(x) + O(xe^{-c\sqrt{\log x}}). \end{aligned}$$

This is the prime number theorem with error term.

9 Recasting the Theory, by Toyesh Jayaswal

The following is taken from chapter 2 of [6].

Some of the properties of the Riemann ζ function can instead be expressed using the theory of p -adic numbers. The real numbers are the completion of $\mathbb{Q}$ under the usual Euclidean metric but for each prime p , a different metric can be chosen for $\mathbb{Q}$ called the p -adic metric. In this metric, numbers that factorize into high powers of p are closer to 0. The completion of $\mathbb{Q}$ under this metric results in the p -adic number fields which have very different properties from the real numbers.

9.1 p -adic numbers

For a rational number r and a prime p , express r as $p^n a/b$ where a, b , and p are all coprime. The value n is called the p -adic order of r and is denoted $\nu_p(r)$. Define the p -adic absolute value as $|r|_p = p^{-\nu_p(r)}$ and $|0|_p = 0$. Because of the number theoretic properties of rational numbers, the p -adic absolute value satisfies the following: $|r|_p = 0$ if and only if $x = 0$ and

$$|r_1 + r_2|_p \leq \max(|r_1|_p, |r_2|_p) \leq |r_1|_p + |r_2|_p.$$

The second property holds because if r_1, r_2 have the same order, then $\nu_p(r_1 + r_2) \geq \nu_p(r_1)$; if they have different orders, then $\nu_p(r_1 + r_2) \geq \min(\nu_p(r_1), \nu_p(r_2))$. From these, the function $d_p(r_1, r_2) = |r_1 - r_2|_p$ can be seen to have the properties of a metric and is known as the p -adic metric. The completion of $\mathbb{Q}$ under this metric is known as the p -adic number field and is denoted $\mathbb{Q}_p$.

9.1.1 Description of the p -adic Numbers

One way to understand the p -adic metric is to look at congruences of integers modulo powers of p . For example, if $p = 5$, then 2 and 27 are equal modulo 5 and 25 but differ modulo 125. Hence, 27 and 2 are 5-adically close on the scale of $1/25$ but far on the scale of $1/125$. Two integers are considered closer together p -adically if they are equal under more congruences modulo p^n . The field $\mathbb{Q}_p$ can also be described in terms of a p -adic number expansion. Let the p -adic integers, $\mathbb{Z}_p$, denote the completion of the integers under the p -adic metric. This can be easily verified to be a ring by the properties of the metric. As mentioned, two integers are considered to be close if they are congruent modulo large p^n but this is equivalent to having the same trailing digits in base p . Hence, the p -adic integers can be represented using an infinite base p expansion that extends to the left.

As an example, consider the expression $x = \dots 444 = \sum_{n=0}^{\infty} 4(5)^n$. Of course, this sum does not converge in the real numbers but the partial sums

$$x_m = \sum_{n=0}^m 4(5)^n = 5^{m+1} - 1$$

differ from -1 by 5^{m+1} . Hence, $|x_m - (-1)|_5 = 5^{-(m+1)} \rightarrow 0$ as $m \rightarrow \infty$ so x converges 5-adically to -1 . As another example, it can be seen that $y = \sum_{n=0}^{\infty} p^n$ converges p -adically to $-1/(p-1)$ but since $|y|_p = 0$, it is a p -adic integer so even numbers in $\mathbb{Q} \setminus \mathbb{Z}$ can be p -adic integers. By the unique prime factorization of rational numbers as $p^n q$ for $|q|_p = 0$, any number in $\mathbb{Q}_p$ can be written with a (possibly infinite to the left) base p expansion. For example, $-1/5$ is 5-adically equal to $-1/5 = \dots 444/5 = \dots 444.4$.

9.1.2 Properties of p -adic Numbers

The following can be found in chapter 6 of [7].

There are many interesting properties of $\mathbb{Q}_p$ that distinguishes it from the topology of the real numbers. Define the open ball $B_r(a) := \{x \in \mathbb{Q}_p \mid |x - a|_p < r\}$. Since $|\cdot|_p$ only takes values of

discrete powers of p , any ball is the countable union of spheres $S_{p^k}(a) := \{x \in \mathbb{Q}_p \mid |x - a|_p = p^k\}$. Hence, $B_{p^k}(a)$ is the same set as the closed ball $\bar{B}_{p^{k-1}}(a)$ and any sphere $S_{p^k}(a) = B_{p^k}(a) \setminus B_{p^{k-1}}(a)$ is an open set. One property that is shared between $\mathbb{Q}_p$ and $\mathbb{R}$ however is that they are both locally compact.

Theorem 9.1. *In $\mathbb{Q}_p$, every closed ball $\bar{B}_r(a)$ is a compact set. In particular, since $\mathbb{Z}_p$ is the closed unit ball centered at 0, $\mathbb{Z}_p$ is compact.*

Now, both $(\mathbb{Q}_p, +)$ and $(\mathbb{Q}_p^\times, \times)$ are locally compact abelian groups. We can now introduce a Haar measure on the group $\mathbb{Q}_p^\times$. A Haar measure on $\mathbb{Q}_p^\times$ is a function $\mu_p^\times$ which maps subsets S of $\mathbb{Q}_p^\times$ to the real numbers satisfying the translational invariance property: $\mu(gS) = \mu(S)$ where $gS = \{gs \mid \forall s \in S\}$. The Haar measure allows us to define an integral over $\mathbb{Q}_p^\times$ in the following manner: letting $S_k = \{x \in S \mid f(x) = k\}$, the integral is defined as

$$\int_S f(x) d\mu_p^\times(x) = \sum_{S_k} k \mu_p^\times(S_k).$$

Some other examples of Haar measures include $d\mu = dx$ for $(\mathbb{R}, +)$ (where translation is taken to be $g+S$) and $d\mu = dx/x$ for $(\mathbb{R}^\times, \times)$. The advantage of the translational invariance can be seen by making a u -substitution $u = ax$ for an integral of the following form

$$\int_S f(ax) \frac{dx}{x} = \int_{aS} f(u) \frac{du}{u}$$

where the constant that comes from changing variables cancels. There is a theorem which states that a Haar measure is uniquely determined up to a multiplicative factor for any locally compact abelian group. By the property

$$\mathbb{Q}_p = \bigcup_{k=-\infty}^{\infty} S_{p^k}(0),$$

and all spheres differ by multiplication with a power of p , it is natural to normalize the Haar measure so that $\mu_p^\times(S_{p^k}(0)) = 1$. With this Haar measure, the following integral arises: let $\varphi(x, s) = |x|_p^s$ for $x \in \mathbb{Z}_p$ and 0 otherwise.

$$\int_{\mathbb{Q}_p^\times} \varphi(x, s) d\mu_p^\times = \sum_{n=0}^{\infty} p^{-s} = \frac{1}{1 - p^{-s}}$$

which is interestingly one of the factors in the Euler product of the Riemann zeta function.

9.2 The Adelic Ring and the Idèle Group

The fields $\mathbb{Q}_p$ and $\mathbb{R}$ form the completions of $\mathbb{Q}$ under different metrics and can be put together to form a mathematical object from which properties of the Riemann zeta function can be obtained.

We start by introducing the Adele ring. Letting $\mathbb{P}$ denote the set of prime numbers, we define the restricted product over $p \in \mathbb{P}$ as the collection of sequences $\{x_p\}_{p \in \mathbb{P} \cup \{\infty\}}$ such that (1) $x_\infty \in \mathbb{R}$ (2) for each $p \in \mathbb{P}$, $x_p \in \mathbb{Q}_p$ and (3) for all but finitely many $p \in \mathbb{P}$, $x_p \in \mathbb{Z}_p$ i.e. for almost all primes, x_p is

p -adic integer. The collections of all such sequences form a ring with operations

$$(x_p) + (y_p) = (x_p + y_p), (x_p) \cdot (y_p) = (x_p \cot y_p)$$

known as the Adele ring of $\mathbb{Q}$, denoted by $\mathbb{A}_{\mathbb{Q}}$. The elements of $\mathbb{A}_{\mathbb{Q}}$ are known as Adeles. The idèle group $\mathbb{I}_{\mathbb{Q}}$ is defined as the collection of all adeles (x_p) such that (1) each component is not zero (2) for all but finitely many $p \in \mathbb{P}$, $x_p \in \mathbb{Z}_p^{\times}$. Here, $\mathbb{Z}_p^{\times}$ denotes the set of p -adic integers whose inverses are also p -adic integers which is all p -adic integers with an absolute value of 1. It is easy to verify that $\mathbb{I}_{\mathbb{Q}}$ forms a commutative group under componentwise multiplication.

André Weil had introduced a topology on $\mathbb{I}_{\mathbb{Q}}$ whose basis around the unit element is of the form

$$U \times \prod_{p \in F} U_p \times \prod_{p \notin F} \mathbb{Z}_p^{\times}$$

where $U \subset \mathbb{R}$ is an open set, $F \subset \mathbb{P}$ is a finite set, and each U_p is an open neighborhood of 1 in $\mathbb{Q}_p$. Weil proved that $\mathbb{I}_{\mathbb{Q}}$ equipped with this topology forms a locally compact Abelian group which implies the existence of a unique (up to a multiplicative constant) Haar measure on it. The Haar measure on $\mathbb{R}^{\times}$ is of the form

$$\mu[(a, b)] = c(\log b - \log a)$$

and as was shown earlier, the Haar measure on each $\mathbb{Q}_p^{\times}$ should take the same value on each $S_{p^k}(0)$. We normalize the Haar measure on $\mathbb{I}_{\mathbb{Q}}$ as follows:

$$\mu \left(\{a \leq t \leq b\} \times \{|x_{p_0}|_{p_0} = p_0^n\} \times \prod_{p \neq p_0} \mathbb{Z}_p^{\times} \right) = 2(\log b - \log a).$$

The properties of $\zeta(s)$ can now be obtained from the idèle group as is outlined in [5]. Define the volume function

$$\|x\| := |x_{\infty}| \times \prod_{p \in \mathbb{P}} |x_p|_p.$$

By definition of the idèle group, almost all of $|x_p|_p$ are 1 so the product ends up being finite. Now define

$$\varphi(x) := \exp(-\pi|x_{\infty}|^2) \prod \mathbf{1}_{\mathbb{Z}_p}(x_p)$$

where $\mathbf{1}_{\mathbb{Z}_p}$ is the indicator function which is 1 on $\mathbb{Z}_p$ and 0 on $\mathbb{Q}_p \setminus \mathbb{Z}_p$. We now define

$$\xi(s) = \int_{\mathbb{I}_{\mathbb{Q}}} \varphi(x) \|x\|^s d\mu(x)$$

The integral can be split into its real and p -adic components so that

$$\xi(s) = \left(\int_{\mathbb{R}^{\times}} e^{-\pi|t|^2} |t|^s \frac{dt}{t} \right) \times \left(\prod_p \int_{\mathbb{Z}_p^{\times}} |x_p|_p^s d\mu_p^{\times}(x_p) \right).$$

The former integral can be evaluated using the u -substitution $u = \pi t^2$

$$\begin{aligned} \int_{\mathbb{R}^\times} e^{-\pi|t|^2} |t|^s \frac{dt}{t} &= 2 \int_0^\infty e^{-\pi t^2} t^{s-1} dt \\ &= 2 \int_0^\infty e^{-u} \left(\frac{u}{\pi}\right)^{\frac{s-1}{2}} \frac{du}{2\sqrt{\pi u}} \\ &= \int_0^\infty e^{-u} \frac{u^{s/2-1}}{\pi^{s/2}} du = \Gamma\left(\frac{s}{2}\right) \pi^{-s/2} \end{aligned}$$

The p -adic integrals were earlier calculated to be $(1 - p^{-s})^{-1}$ so putting everything together,

$$\xi(s) = \Gamma\left(\frac{s}{2}\right) \pi^{-s/2} \prod_p \frac{1}{1 - p^{-s}} = \Gamma\left(\frac{s}{2}\right) \zeta(s) \pi^{-s/2}$$

which shows that this $\xi(s)$ is the same one seen earlier. The functional equation for $\zeta(s)$ can also be obtained by evaluating the integral in an alternate fashion. The rational numbers embed naturally into $\mathbb{I}_{\mathbb{Q}}$ sending the rational number α to the constant series (α) which are called the group of principal idèles. Since any rational number has only finitely many prime divisors, it is a unit in almost all $\mathbb{Z}_p^\times$ so all the conditions of being an idèle are met. The integral above can now be split into this principal group and multiples of it. We define the quotient group $C_{\mathbb{Q}} := \mathbb{I}_{\mathbb{Q}}/\mathbb{Q}^\times$ as the set of all equivalence classes $\bar{x} = \{\alpha x \mid \alpha \in \mathbb{Q}\}$. Intuitively, each of these sets can be assigned a representative so that each $x \in \mathbb{I}_{\mathbb{Q}}$ can be expressed as the product of some representative in $C_{\mathbb{Q}}$ and a principal idèle. Some operations on idèles can be transferred over to $C_{\mathbb{Q}}$. Multiplication can be defined as

$$\bar{x} \cdot \bar{y} = \overline{xy} = \{\alpha xy \mid \alpha \in \mathbb{Q}\}$$

and it can be checked that it is independent of the representatives x, y chosen. The volume function can also be transferred over to $C_{\mathbb{Q}}$. Consider the principal idèle (α) . If $\alpha = \prod p_i^{k_i}$,

$$\|(\alpha)\| = |\alpha|_\infty \prod_p |\alpha|_p = \prod_p p_i^{k_i} \prod p_i^{-k_i} = 1.$$

The generalized version of this theorem is known as the Artin-Whaples theorem. Moreover, it follows from the properties of norms that the volume function is multiplicative. Hence, for some idèle x and some principal idèle (α) , $\|(\alpha)x\| = \|(\alpha)\| \|x\| = \|x\|$. Hence, any representative of a set in $C_{\mathbb{Q}}$ has the same volume function and it makes sense to write $\|\bar{x}\| = \|x\|$ for $x \in \bar{x} \in C_{\mathbb{Q}}$.

Skipping over the measure theoretic details, the integral for $\xi(s)$ can be evaluated as

$$\xi(s) = \int_{\mathbb{Q}^\times} \left(\int_{C_{\mathbb{Q}}} \varphi(\alpha x) \|\bar{x}\|^s d\bar{\mu}(\bar{x}) \right) d\chi(\alpha),$$

where $\bar{\mu}$ is known as the induced measure on $C_{\mathbb{Q}}$ and χ is the counting measure on $\mathbb{Q}^\times$ since it is

discrete. The outer integral can be evaluated as a sum and

$$\xi(s) = \int_{C_{\mathbb{Q}}} \left(\sum_{\alpha \in \mathbb{Q}^{\times}} \varphi(\alpha x) \right) \|\bar{x}\|^s d\bar{\mu}(\bar{x}).$$

It will now be shown that $\sum_{\alpha \in \mathbb{Q}^{\times}} \varphi(\alpha x)$ can in fact be written as a function in $C_{\mathbb{Q}}$ and does not depend on the choice of representative x . Define

$$\Theta(\bar{x}) = 1 + \sum_{\alpha \in \mathbb{Q}^{\times}} \varphi(\alpha x)$$

but by the definition of φ , the sum is only taken over values for which αx is in $\mathbb{Z}_p$ for all p or in other words, the α for which

$$p^{\nu_p(\alpha)} \geq |x_p|_p$$

for all p . Now, by factorizing α as $\pm \prod_p p^{\nu_p(\alpha)}$, the factor coming from $\exp(-\pi|x_{\infty}|^2)$ becomes

$$\begin{aligned} \alpha^2 |x_{\infty}|^2 &= \left(\pm \prod_p p^{\nu_p(\alpha)} |x_p|_p^{-1} |x_p|_p \right)^2 |x_{\infty}|^2 \\ &= \left(\prod_p p^{\nu_p(\alpha)} |x_p|_p^{-1} \right)^2 \|x\|^2. \end{aligned}$$

By the earlier condition, the product $\prod_p p^{\nu_p(\alpha)} |x_p|_p^{-1}$ must be a natural number and as α varies over $\mathbb{Q}^{\times}$, each number will be attained exactly twice (since $\pm \alpha$ gives the same natural number). Hence,

$$\Theta(\bar{x}) = 1 + \sum_{\alpha \in \mathbb{Q}^{\times}} \varphi(\alpha x) = 1 + 2 \sum_{n=1}^{\infty} e^{-\pi n^2 \|\bar{x}\|^2}.$$

By the Poisson summation formula (2), it follows that $\Theta(\bar{x}) = \|\bar{x}\|^{-1} \Theta(\bar{x}^{-1})$. The integral is then split into two parts: one for $\|\bar{x}\| \leq 1$ and the other for $\|\bar{x}\| > 1$. The Haar measure is left invariant under $\bar{x} \rightarrow \bar{x}^{-1}$, so

$$\begin{aligned} \xi(s) &= \int_{\|\bar{x}\| \leq 1} + \int_{\|\bar{x}\| > 1} (\Theta(\bar{x}) - 1) \|\bar{x}\|^s d\bar{\mu}(\bar{x}) \\ &= \int_{\|\bar{x}\| > 1} (\|\bar{x}\|^s + \|\bar{x}\|^{1-s}) (\Theta(\bar{x}) - 1) d\bar{\mu}(\bar{x}) + \int_{\|\bar{x}\| > 1} (\|\bar{x}\|^{1-s} - \|\bar{x}\|^{-s}) d\bar{\mu}(\bar{x}). \end{aligned}$$

Both integrals are left invariant under switching s and $1 - s$ giving the functional equation $\xi(s) = \xi(1 - s)$ from which the functional equation for the ζ function can be deduced. Furthermore, the second integral evaluates to $\frac{1}{s(s-1)}$ so $\xi(s)$ is meromorphic with poles at 0, 1. Using this theory, the properties of the ζ function discussed earlier can be derived, but it also provides the advantage of generalizing to arbitrary number fields.

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